10	At a point on the surface of a uniform sphere of diameter *d*, the gravitational field strength due to the sphere is *X*. What would be the gravitational field strength on the surface of a uniform sphere of the same density but of diameter 3*d*?							
	A	2*X*	**B**	3*X*	**C**	5*X*	**D**	8*X*

Answer: B

$\boldsymbol{M = density \text{ x } volume = \rho\left(\frac{4}{3}\pi R^3\right)}$ where *M* is the mass, ρ the density and *R* the radius of sphere.

Gravitational field strength at the surface, $\boldsymbol{g = \frac{GM}{R^2} = \frac{G4\pi R^3\rho}{3R^2} = \frac{G4\pi R\rho}{3} = \frac{G4\pi d\rho}{6}}$ where *d* is the diameter of sphere.

Since *g* is directly proportional to *d*, the gravitational field strength due to the sphere of diameter 3*d* will be 3*X*.

11	Which of the following statement about a geostationary satellite around Earth is *true*?